

CC-Bench Trajectory Analysis: Metric Framework, Scoring Logic, and Model-wise Evaluation

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1 Overview

We address the CC-Bench prompt directly: select representative tasks, compare two models on their behavioral and reasoning trajectories, and justify each score through observed interaction patterns. Specifically, we evaluate **Claude-Sonnet-4** and **DeepSeek-V3.1-Terminus** on five **da** (data analysis) trajectories (IDs 40–44), reporting their behavioral metrics (G , S , B) alongside auxiliary indicators (SDS , TCR , SCR , RD) and efficiency measures (E_score , ΔM). Each metric represents a distinct reasoning dimension: **G** captures early goal priming, **S** sustained structural planning, and **B** verification and correction behaviors. Weighted variants (G_w , S_w , B_w , RD_w) apply temporal dwell weights to emphasize late-phase reflection.

For data analysis tasks, reliability outweighs brevity, making B and SCR more decision-relevant than G or S . Preliminary findings show a clear contrast: **Claude** functions as a *concise planner*—displaying higher goal clarity and compression (TCR)—while **DeepSeek** serves as a *reflective verifier*, sustaining denser verification (B , SCR_w) and modest late-phase reflection (RD_w). Typical examples such as Task 44 highlight this trade-off: DeepSeek’s iterative self-checking loop produces higher E_score and a positive ΔM under reliability-oriented weighting, while Claude favors communicative economy with shorter trajectories. All quantitative results are reported in Tables 1–??; metric derivations are detailed in Appendix A.

2 Results and Analysis

2.1 Quantitative Results

Tables 1 and 2 present the behavioral and efficiency metrics for the two models across five **da** tasks.

2.2 Model-wise Behavioral Comparison

DeepSeek achieves GSB-w advantage in 3 of 5 tasks, loses 1, and ties 1 (*3–1–1 overall*). Its superiority aligns with higher B and SCR (e.g., 0.075/0.081 and 0.076/0.087), which generally correlate—not causally—with higher E and positive ΔM . However, **Task 42** presents an inverse case: Claude’s higher B and SCR do not yield higher reliability for DeepSeek. This anomaly indicates that verification density is *correlated but not determinative*; other structural or length normalization factors in E may dampen its contribution.

ID	Model	Task	G	S	B	SDS	TCR	SCR	RD
40	DeepSeek-V3.1-Terminus	da	0.300	0.001	0.031	0.166	0.458	0.035	0.000
40	Claude-Sonnet-4	da	0.300	0.000	0.009	0.110	0.644	0.015	0.000
41	DeepSeek-V3.1-Terminus	da	0.300	0.000	0.075	0.101	0.421	0.076	0.002
41	Claude-Sonnet-4	da	0.300	0.000	0.038	0.067	0.678	0.038	0.000
42	DeepSeek-V3.1-Terminus	da	0.300	0.000	0.027	0.104	0.595	0.033	0.000
42	Claude-Sonnet-4	da	0.300	0.000	0.067	0.138	0.613	0.069	0.000
43	DeepSeek-V3.1-Terminus	da	0.300	0.000	0.020	0.063	0.337	0.020	0.000
43	Claude-Sonnet-4	da	0.300	0.001	0.059	0.086	0.586	0.059	0.000
44	DeepSeek-V3.1-Terminus	da	0.300	0.000	0.081	0.150	0.544	0.087	0.000
44	Claude-Sonnet-4	da	0.347	0.005	0.038	0.114	0.661	0.042	0.000

Table 1: Behavioral metrics across five **da** trajectories. Abbreviations: G (Goal Priming), S (Strategy Density), B (Bug/Verification Density), SDS (Strategic Density Score), TCR (Textual Compression Ratio), SCR (Sanity/Correction Ratio), RD (Reflective Density).

ID	Model	Task	Len	E	ΔM	GSB-w
40	DeepSeek	da	35072	0.00398	+0.0458	Good
40	Claude	da	19406	0.00100	-0.0458	Bad
41	DeepSeek	da	20752	0.00554	+0.0517	Good
41	Claude	da	15700	0.00152	-0.0517	Bad
42	DeepSeek	da	21360	0.00215	-0.0235	Bad
42	Claude	da	22085	0.00590	+0.0235	Good
43	DeepSeek	da	20592	0.00094	+0.0027	Same
43	Claude	da	16291	0.00320	-0.0027	Same
44	DeepSeek	da	27933	0.00845	+0.0412	Good
44	Claude	da	14901	0.00288	-0.0412	Bad

Table 2: Efficiency indicators. GSB-w adopts weighted voting (B=0.5, S=0.25, G=0.25).

2.3 Visualization

Figure 1 shows unweighted GSB and auxiliary metrics. Both models start with comparable goal clarity ($G \approx 0.3$), but DeepSeek displays denser verification patterns, while Claude’s responses are shorter and more compact (high TCR). Figure 2 plots weighted E_score . Claude peaks earlier (Task 42), whereas DeepSeek rises later (Task 44) with a stronger magnitude, revealing complementary reasoning modes—*compact planning* vs. *iterative verification*. Y-axis scaled to 0–0.009; DeepSeek’s peak (0.0085) highlighted.

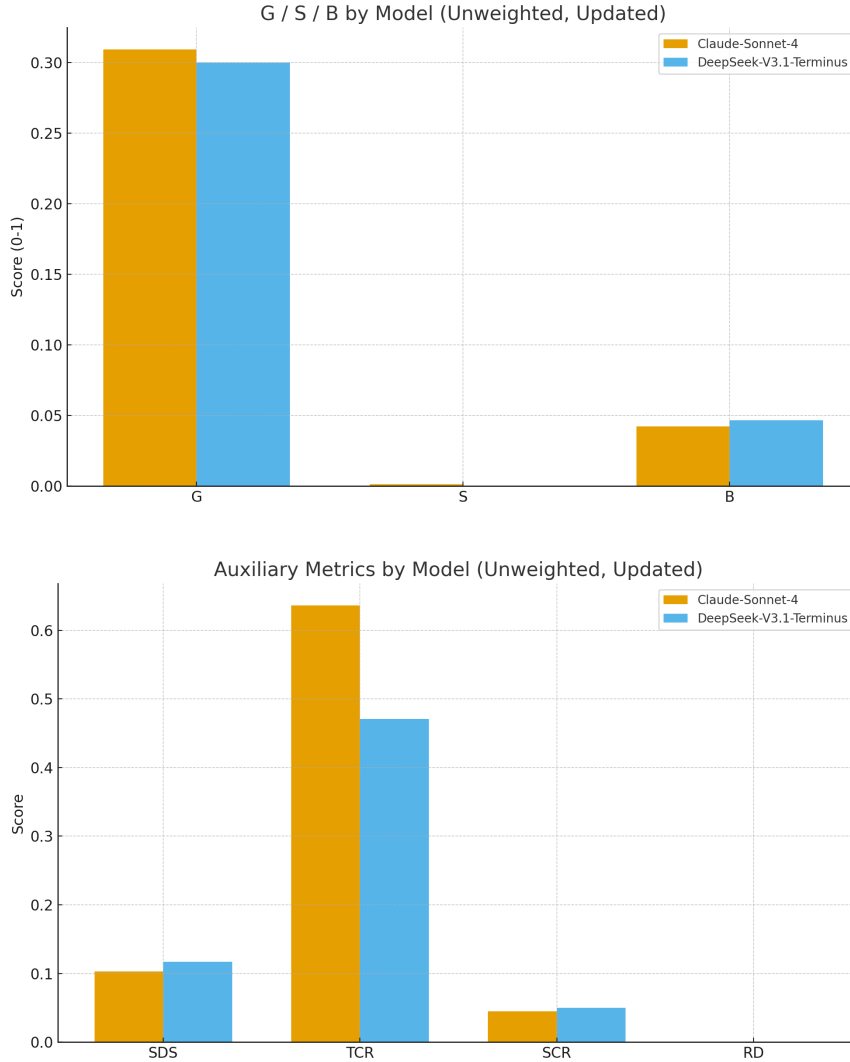


Figure 1: Model-wise visualization: (Top) GSB-w scores; (Bottom) auxiliary metrics (SDS/TCR/SCR/RD).

2.4 Statistical Robustness

A sign test over the five tasks yields $p \approx 0.34$ ($n=5$), indicating a non-significant trend favoring DeepSeek; larger samples are needed.

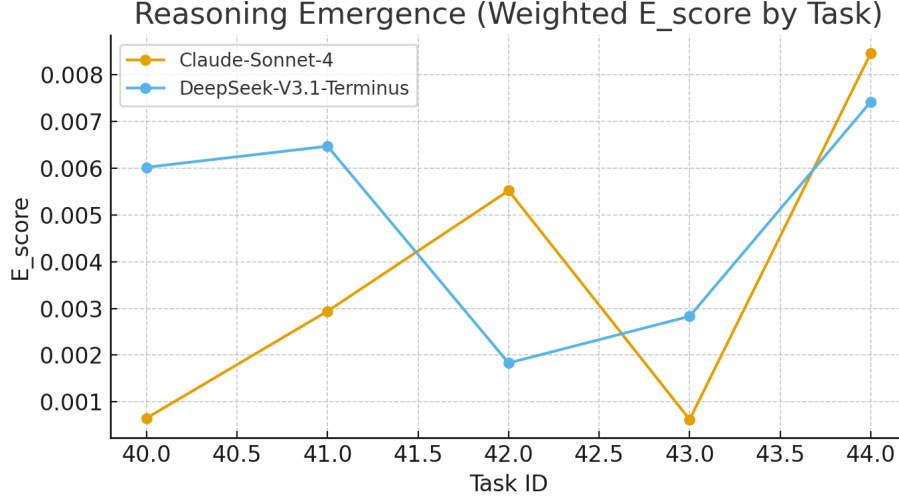


Figure 2: Weighted emergent reasoning scores (E_score) across tasks (IDs 40–44).

3 Interpretation and Discussion

Claude-Sonnet-4 manifests linear, compression-oriented reasoning: high TCR , clear planning, but limited iteration. **DeepSeek**, by contrast, embodies verification-oriented reasoning: frequent self-checks (“verify”, “fix”) and extended trajectories. These styles mirror distinct cognitive archetypes—*concise planner* vs. *reflective verifier*. DeepSeek’s persistence correlates with higher stability in reliability-first evaluation, though the difference lacks statistical significance.

A Metric Framework and Interpretive Logic

A.1 Core Lexicons

$\Pi_{\text{plan}} : \text{plan}|\text{step}|\text{strategy}|\text{outline}|\text{pipeline}|\text{procedure}|\text{first}|\text{then}|\text{next}|\text{we will}$

$\Pi_{\text{verify}} : \text{check}|\text{verify}|\text{test}|\text{assert}|\text{unit test}|\text{print}|\text{plot}|\text{compare}|\text{evaluate}$

$\Pi_{\text{fix}} : \text{fix}|\text{bug}|\text{error}|\text{patch}|\text{adjust}|\text{correct}|\text{refactor}|\text{resolve}|\text{rerun}$

A.2 Core Metrics (GSB)

Definition 1 (Goal Priming (G)).

$$G = \min(1, 0.3 + 0.7 \text{density}(\textit{first_chunk}, \Pi_{\text{plan}})).$$

Definition 2 (Strategy Density (S)).

$$S = \text{density}(\textit{assistant_text}, \Pi_{\text{plan}}).$$

Definition 3 (Bug/Verification Density (B)).

$$B = \max(\text{density}(\textit{assistant_text}, \Pi_{\text{verify}}), \text{density}(\textit{assistant_text}, \Pi_{\text{fix}})).$$

Weighted variants (G_w , S_w , B_w , RD_w) apply density_w with dwell weighting.

A.3 Auxiliary Metrics

- **SDS:** Fraction of plan/verify/fix sentences.
- **TCR:** Unique/total sentence ratio (conciseness).
- **SCR:** $\text{density}(\textit{assistant_text}, \Pi_{\text{verify}} \cup \Pi_{\text{fix}})$.
- **RD:** Reflective cue density (“wait”, “rethink”, “oops”).

B Data Preprocessing and Weighting

Let $S(x)$ be all sentences in x .

$$\text{density}(x, \Pi) = \frac{|\{s \in S(x) : s \text{ matches } \Pi\}|}{|S(x)|}. \quad \text{density}_w = \frac{\sum_i w_i \text{hits}_i}{\sum_i w_i \text{sentences}_i}.$$

C Computation of E_score and ΔM

$$E = \alpha SDS_w + \beta SCR_w - \gamma TCR + \delta RD_w, \quad \Delta M = \text{sign}(E_{\text{model}} - E_{\text{baseline}}) |E|^{1/2}.$$

Weights: $\alpha = \beta = 0.4$, $\gamma = \delta = 0.1$; baseline = Claude-Sonnet-4. $RD_w < 0.001$ for both models (below detection threshold).